

DEPARTMENT OF ELECTRONICS ENGINEERING

ELEC366
COMMUNICATIONS LABORATORY

EXPERİMENT-5

1. GOAL

In this experiment, you will learn how to perform amplitude shift keying (ASK) modulation & demodulation and frequency shift keying (FSK) modulation & demodulation.

2. BACKGROUND

2.1 Amplitude Shift Keying Modulation

An essential part of electronic communications and telecommunications is the ability to share the channel. This is true regardless of whether the channel is copper wire, optical fiber or free-space. If it's not shared then there can only ever be one person transmitting on it at a time. Think about the implications of this for a moment. Without the ability to share, there could only be one radio or TV station in each area. Only one mobile phone owner could use their phone in each cell at any one time. And there would only be the same number of phone calls between any two cities as the number of copper wires or optical fibers that connected them.

So sharing the channel is essential and there are several methods of doing so. One is called time division multiplexing (TDM) and involves giving the users exclusive access to the channel for short periods of time. On the face of it, this type of sharing might seem impractical. Imagine giving all mobile phone users in a cell just a minute or so to make their call then having to wait until their turn comes around again. However, TDM works well when the access time is extremely short (less than a second) and the rate of the sharing is fast. This allows multiple users to appear to have access all at the same time.

TDM is used for digital communications and is achieved by interleaving the users' data. That is, a portion of one user's data is transmitted followed by a portion of the next user's data and so on. Unfortunately, there's a catch. If the message is real-time information that cannot afford to be delayed (like digitally encoded speech) then, as the number of users increases, so must the data's bit-rate.

Another method of sharing the channel is called frequency division multiplexing (FDM) and involves giving the users exclusive and uninterrupted access to a portion of the channel's radio frequency spectrum. To transmit their message the user must superimpose it onto a carrier that sits inside their allocated band of frequencies. This method is used by broadcast radio and television to share free-space.

FDM is also used for digital communications and uses the same modulation schemes available to analog communications including: AM, DSB-SC and FM. When AM is used for

multiplexing digital data, it is known as amplitude shift keying (ASK). Other names include: on-off keying, continuous wave and interrupted continuous wave.

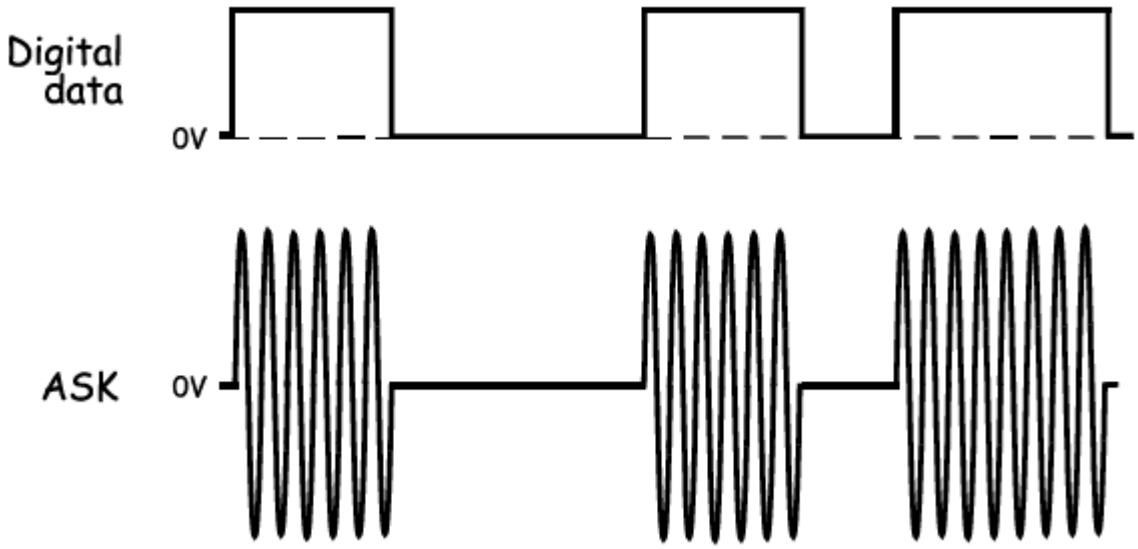


Figure 8.1: Digital data and ASK signal

Notice that the ASK signal's upper and lower limits (the envelopes) are the same shape as the data stream (though the lower envelope is inverted). This is simultaneously an advantage and a disadvantage of ASK. Recovery of the data stream can be implemented using a simple envelope detector. However, noise on the channel can change the envelopes' shape enough for the receiver to interpret the logic levels incorrectly causing errors (analog AM communications have the same problem and the errors are heard as a hiss, crackles and pops).

Here you'll examine the operation of a method that involves using the digital signal to switch the carrier's connection to the channel on and off.

In digital modulation, frequency, amplitude and phase of a signal are used to modulate a digital input. Amplitude Shift Keying (ASK) is a digital modulation technique. Pulse Amplitude Modulation is also known as ASK modulation. In this way, modulated signals can have discrete values in amplitude for a given digital input and it is constant for a symbol period. Modulation process is described in Figure 8.1 and Figure 8.2.

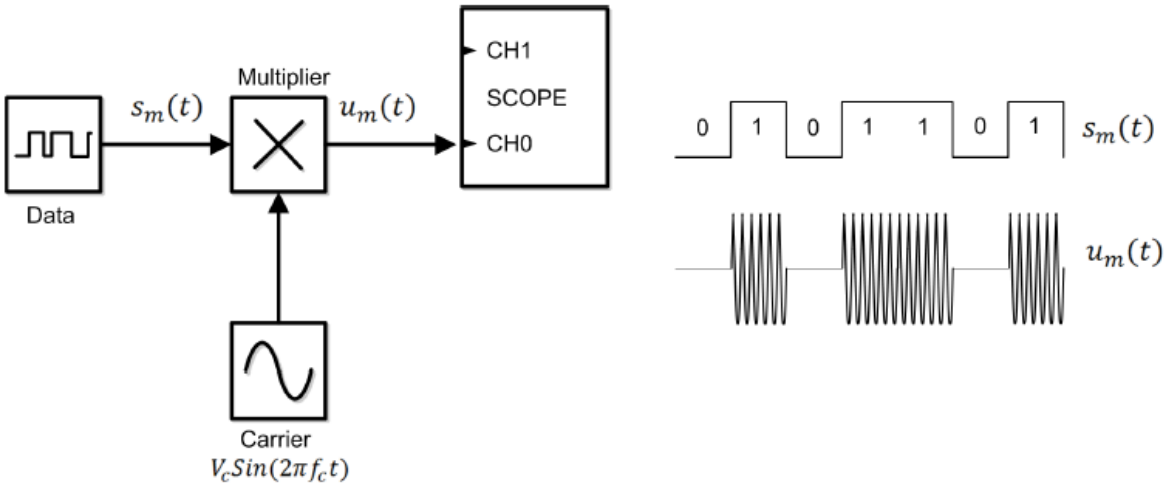


Figure 8.2: ASK modulator block diagram and an ASK signal

Alternatively, non-binary ASK modulation is also available to transmit more than one bits using an ASK signal. An example of a non-binary ASK signal is provided in Figure 8.3.

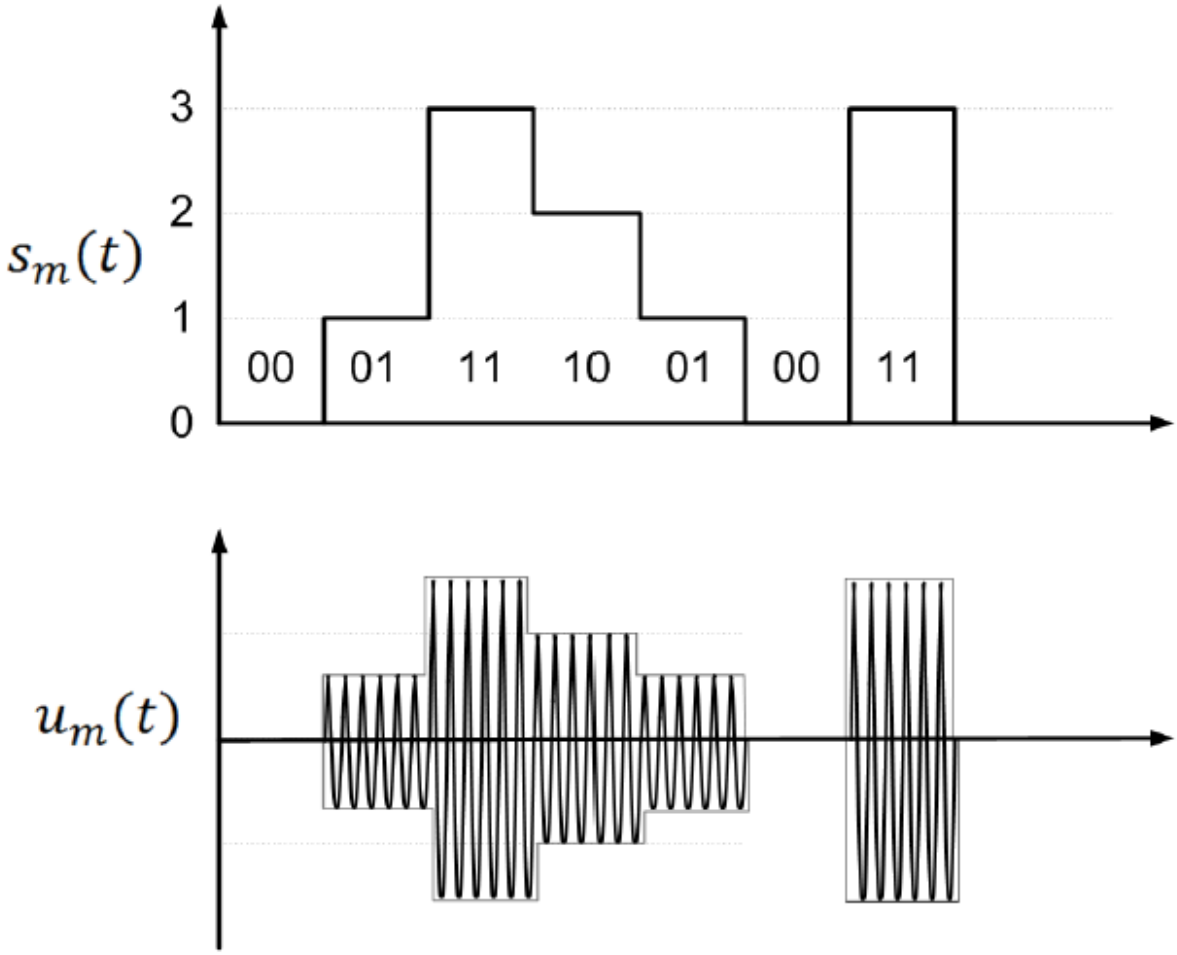


Figure 8.3: Non-binary ASK signals

2.2 Amplitude Shift Keying Demodulation

Demodulation process of ASK signals can be seen as a detection problem of an amplitude of the received ASK signals. In this way, digital input signal controls amplitude of the modulated signal. Implementation of the process is shown in Figure 8.4.

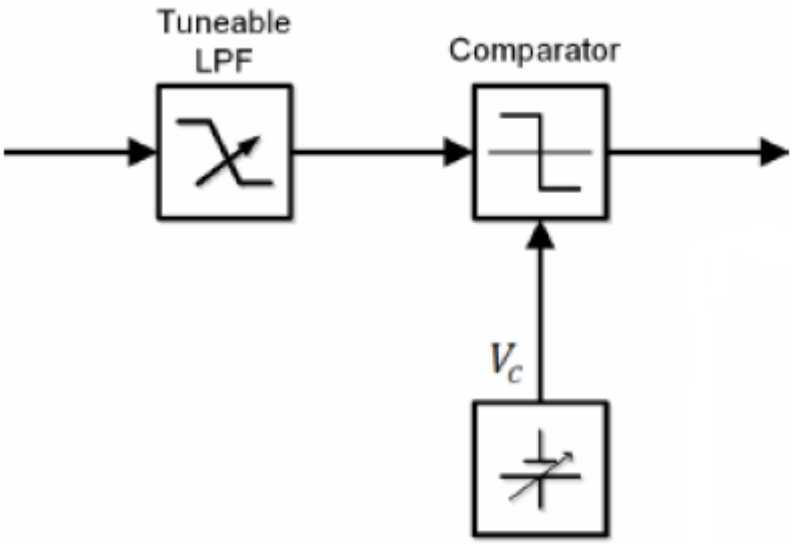


Figure 8.4: Demodulation of an ASK signal

2.3. Frequency Shift Keying Modulation

Frequency division multiplexing (FDM) allows a channel to be shared among a set of users. Recall that this is achieved by superimposing the message onto a carrier signal inside the user's allocated portion of the radio-frequency spectrum. Recall also that any of the analog modulation schemes can be used to transmit digital data in this way. When frequency modulation (FM) is used it is known as binary frequency shift keying (BFSK or more commonly just FSK).

One of the reasons for using FSK is to take advantage of the relative noise immunity that FM enjoys over AM. Recall that noise manifests itself as variations in the transmitted signal's amplitude. These variations can be removed by FM/FSK receivers (by a circuit called a limiter) without adversely affecting the recovered message.

Figure 8.5 shows what an FSK signal looks like time-coincident with the digital signal that has been used to generate it.

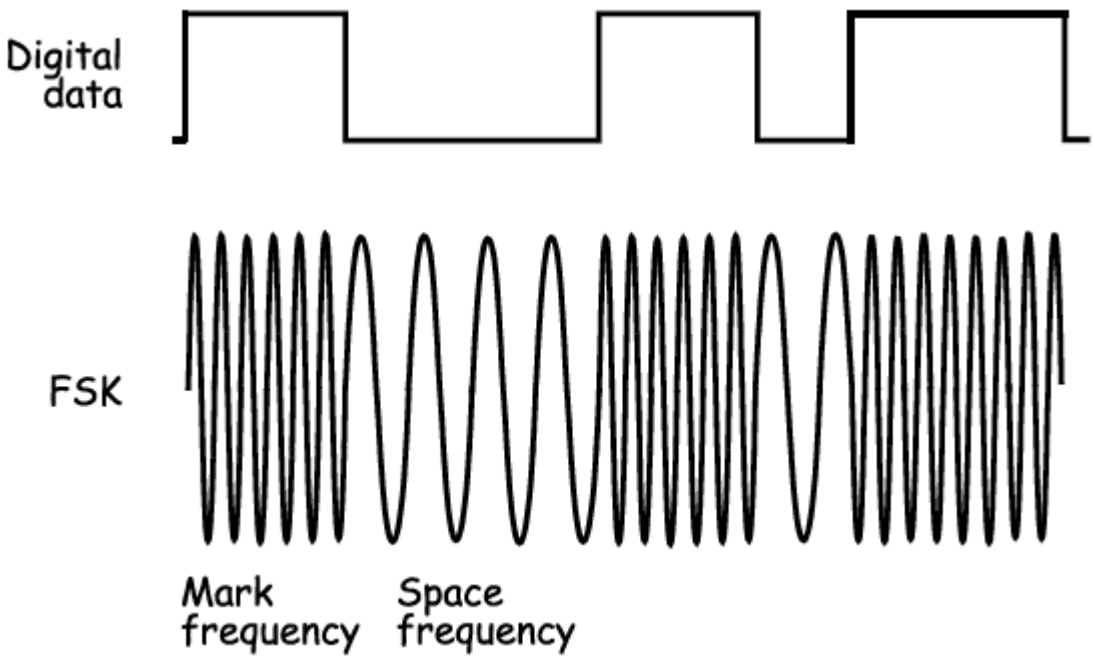


Figure 8.5: Typical FSK signal

Notice that the FSK signal switches between two frequencies. The frequency of the signal that corresponds with logic-0s in the digital data (called the space frequency) is usually lower than the modulator's nominal carrier frequency. The frequency of the signal that corresponds with logic-1s in the digital data (called the mark frequency) is usually higher than the modulator's nominal carrier frequency. The modulator doesn't output a signal at the carrier frequency, hence the reference here to it as being the "nominal" carrier frequency.

FSK generation can be handled by conventional FM modulator circuits and the voltage-controlled oscillator (VCO) is commonly used. Similarly, FSK demodulation can be handled by conventional FM demodulators such as the zero-crossing detector and the phase-locked loop. Alternatively, if the FSK signal is passed through a sufficiently selective filter, the two sinewaves that make it up can be individually picked out. Considered on their own, each signal is an ASK signal and so the data can be recovered by passing either one of them through an envelope detector.

Frequency Shift Keying (FSK) is a digital modulation technique. In this way, modulated signals can have discrete frequencies for a given digital input and it is constant for a symbol period. Modulation process is described in Figure 8.6.

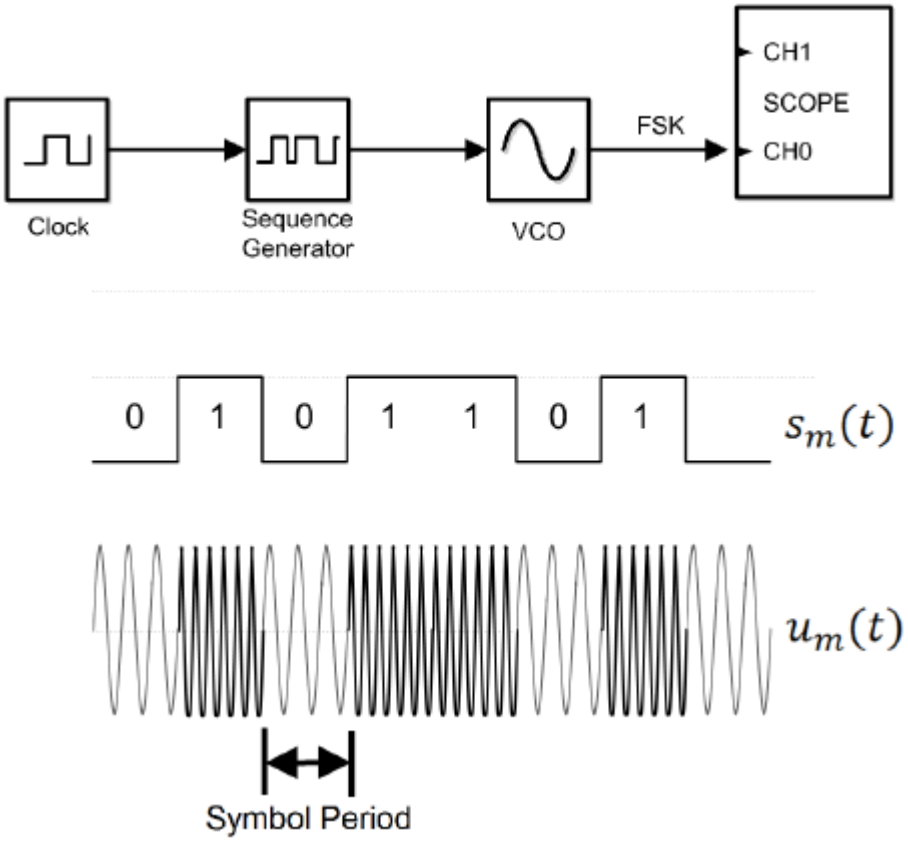


Figure 8.6: FSK modulator block diagram and a FSK signal

2.4. Frequency Shift Keying Demodulation

Demodulation process for an FSK signal can be simply implemented by the following block diagram.

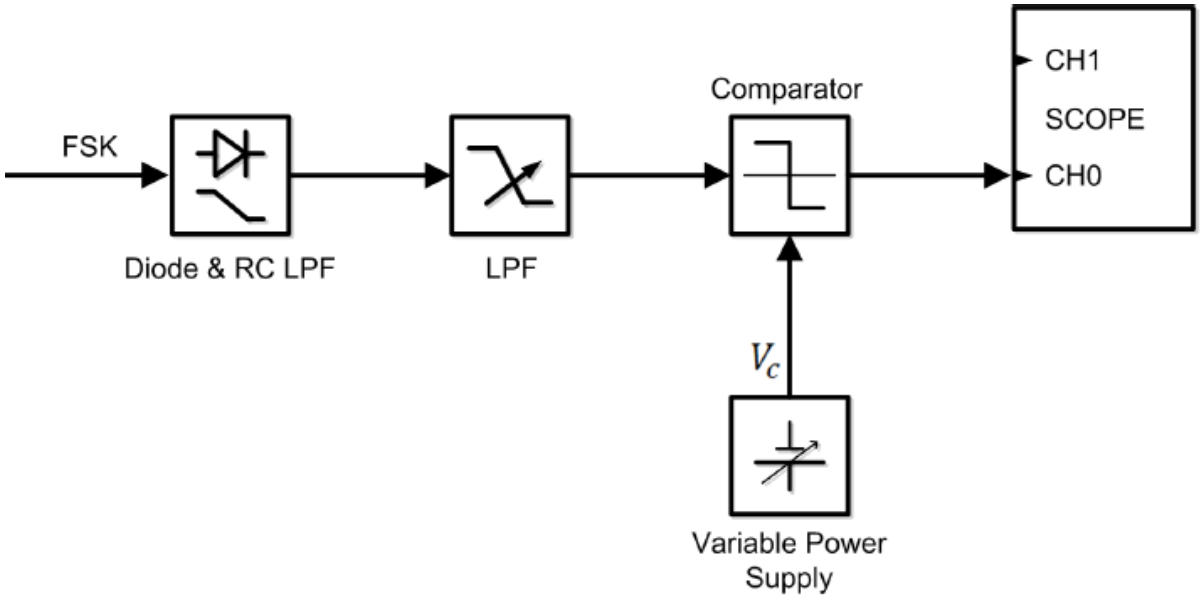


Figure 8.7: FSK demodulator block diagram

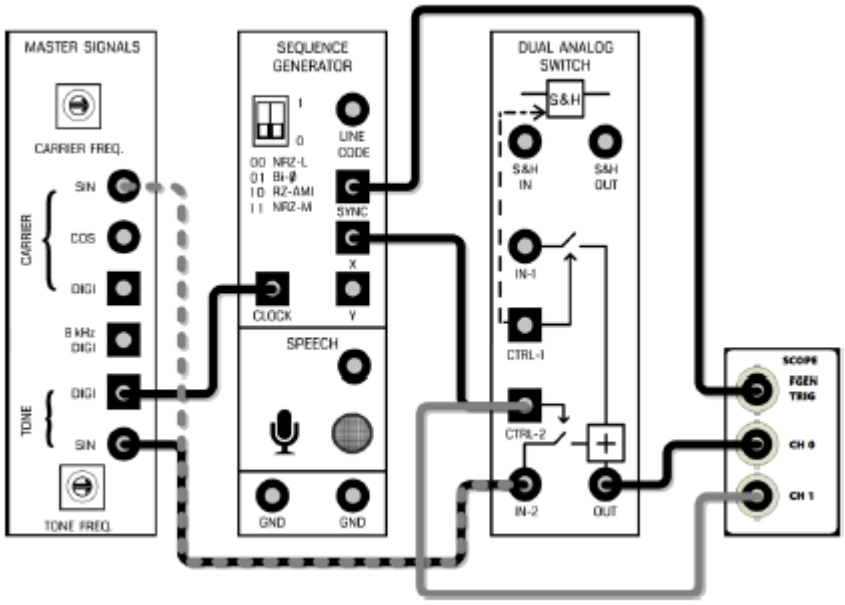
FSK modulation is a robust technique for distortion of the amplitude.

3. PROCEDURE

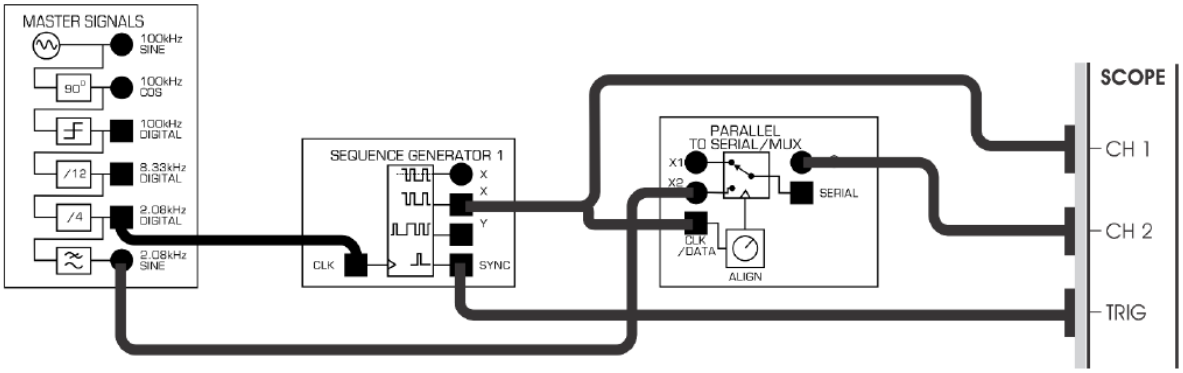
3.1. Generating an ASK signal

Red Set	Blue Set
• Please turn on the power of TETRA-BB and ELVIS platform. Then, connect the ELVIS platform to PC (via USB). • Set the control mode of TETRA-BB to PC. • Run the TETRA-BB soft front panel (SFP). After selecting the "Exp. 8 - ASK Modulation & Demodulation" experiment from the menu, TETRA-BB software will provide guidance for the steps of implementing ASK modulation, whose block diagram and the required TETRA-BB	▪ Ensure that the NI ELVIS III Application Board power button at the top left corner of the unit is OFF (not illuminated). ▪ Carefully plug the Emona Communications board into the NI ELVIS III ensuring that it is fully engaged both front and back. ▪ Ensure that you have connected the NI ELVIS III to the PC using the USB cable and that the PC is turned on. ▪ Turn on the Application Board *Power* button by pressing it once and confirm that it is illuminated. The LEDs on the board should also be illuminated. If

connections are shown in Figure 8.8.a (steps 0-6 in the TETRA-BB SFP).	they are not, then switch the unit off immediately and check for connection or insertion errors. ▪ Open the Instrument Launcher software in your browser and select the required instruments. ▪ Modify the set-up as shown in figure 8.8.b.
• Set the frequency of sinus tone signal. Frequency (tone) = 2 kHz • Adjust following the oscilloscope panel: Channel 0 Voltage Range: 1V/div Channel 1 Voltage Range: 2V/div Horizontal Time base: 1ms/div Trigger: Digital, TRIG Source, Level:0V Horizontal Position (%):50 Probe Attenuation :1x	▪ Adjust following Scope configurations: Channel Voltage Range: 1V/div Horizontal Time base: 500us/div Trigger: Digital Edge, Source: TRIG input, Rising Probe Attenuation :1x



(a)



(b)

Figure 8.8. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to generate an ASK signal.

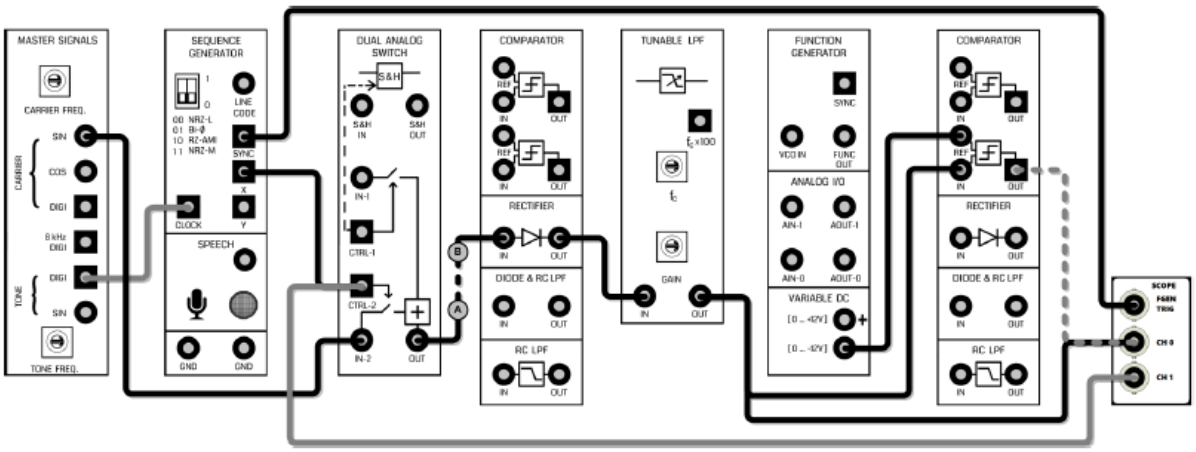
1. Observe Observe and draw in time domain output of the sequence generator and OUT port of the dual analog switch block (red set) or parallel to serial / mux block (blue set). 📷,✏
2. Disconnect 2 kHz tone signal from dual analog switch block (red set) or parallel to serial / mux block (blue set). Connect 100 kHz sinus signal to IN 2 port (red set) or X2 port (blue set). Observe in time domain output of the sequence generator and OUT port of the dual analog switch block (red set) or parallel to serial / mux block (blue set). 📷,✏
3. Stop oscilloscope. Switch from oscilloscope to DSA in TETRA-BB SFP (red set) or enable the FFT mode of the Oscilloscope instrument (blue set). Adjust DSA and FFT mode settings as given below. Observe the frequency spectrum of ASK modulated signal. 📷,✏

Red Set	Blue Set
• Stop oscilloscope. • Switch from the oscilloscope panel to the DSA panel. • Apply the settings of DSA given below: Input Settings: Channel 0 Center Frequency: 100000 Frequency Span: 20000 Mode: RMS Resolution (lines): 3200 Weighting: Exponential	▪ Enable the FFT mode of the Oscilloscope instrument. Change the scope's timebase to 2V/div. ▪ Set the frequency span for the FFT displayed from say 0kHz to 20kHz for a closeup of the frequency domain of interest. ▪ Select the Hamming window.

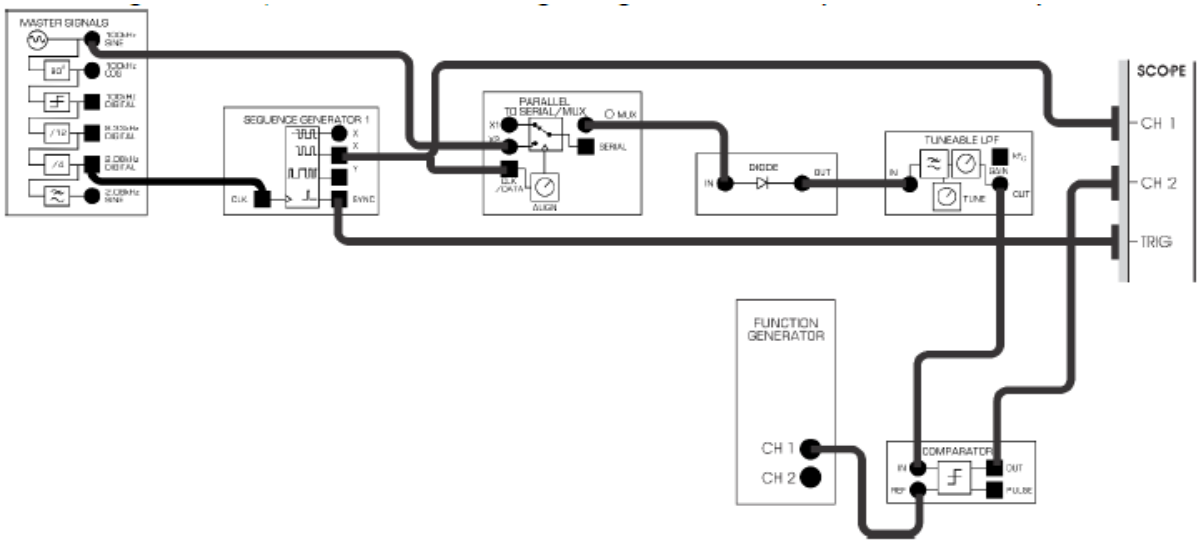
Window: 7 Term-B Harris # of Averages: 5	

3.2. Demodulating an ASK Signal

Red Set	Blue Set
• Select the "Exp. 8 - ASK Modulation & Demodulation" experiment from the menu, TETRA-BB software will provide guidance for the steps of implementing demodulation of an ASK signal, whose block diagram and the required TETRA-BB connections are shown in Figure 8.9.a (steps 7-12 in the TETRA-BB SFP). • Adjust following the oscilloscope panel: Channel 0 Voltage Range: 1V/div Channel 1 Voltage Range: 5V/div Horizontal Time base: 1ms/div Trigger: Digital, TRIG Source, Level: 0V Horizontal Position(%): 50	▪ Modify the set-up as shown in figure 8.9.b. ▪ Adjust following Scope configurations: Channel Voltage Range: 1V/div Horizontal Time base: 500us/div Trigger: Digital Edge, Source: TRIG input, Rising Probe Attenuation :1x ▪ Use the ELVIS III Function Generator output Channel 1 to create a DC voltage of about 1V and 1 kS/s by loading the Custom waveform file "ECB_positive1V_DC.csv".



(a)



(b)

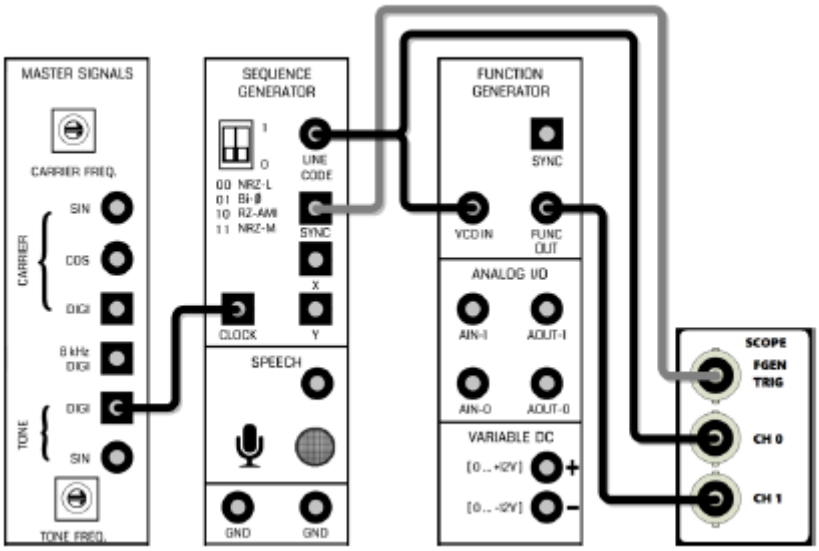
Figure 8.9. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to demodulate an ASK signal.

4. Adjust the cut-off frequency of tunable low pass filter by the use of oscilloscope to observe output of the filter a square waveform. Then draw signals at the input and the output of the low pass filter 📷,🖊.
5. Connect the output of the low pass filter to comparator block. Adjust negative DC voltage (red set) or create a DC voltage of about 1V and 1 kS/s from func. gen. (blue set). Observe and draw the output of the comparator block and the input signal at Control 2 port of the Dual analog switch (red set) or input signal at parallel to serial / mux block (blue set) in time domain 📷,🖊.

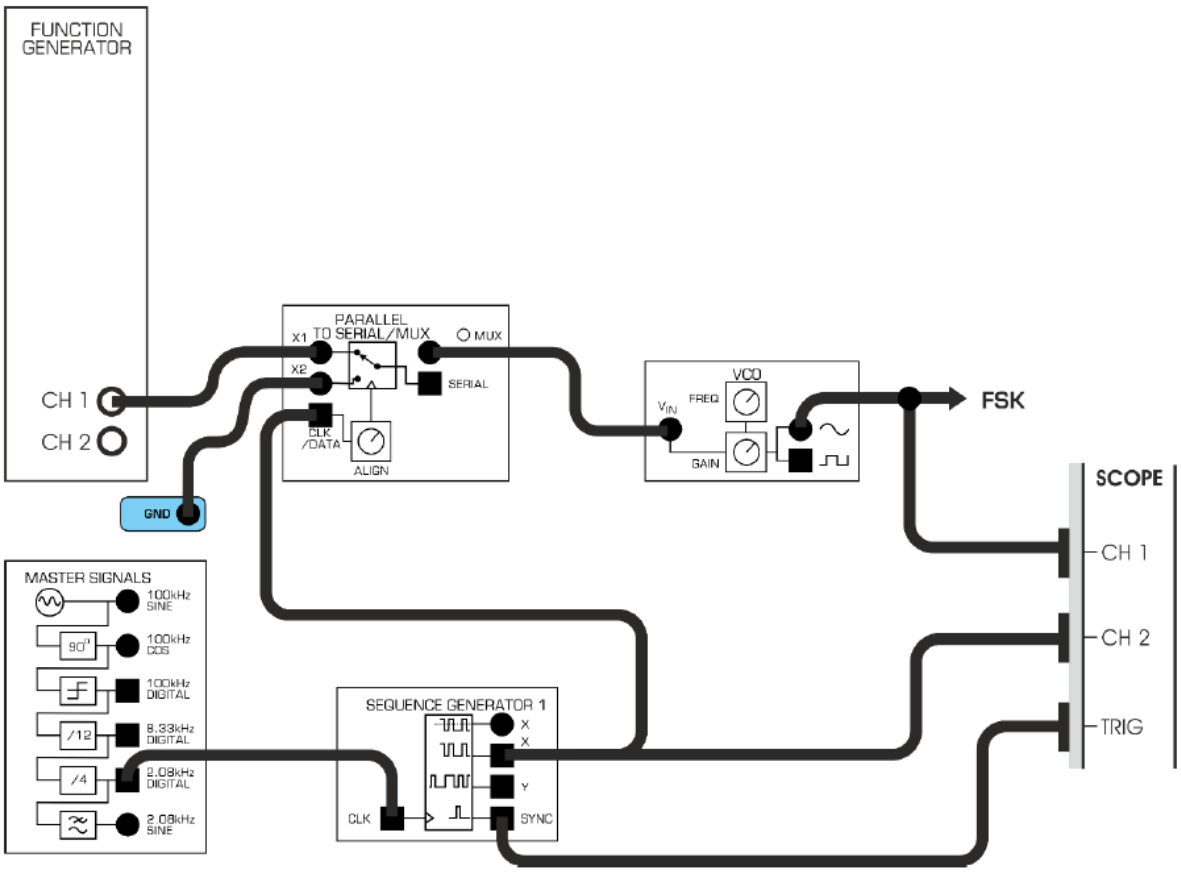
3.3. Generating an FSK signal

Red Set	Blue Set
• Please turn on the power of TETRA-BB and ELVIS platform. Then, connect the ELVIS platform to PC (via USB). • Set the control mode of TETRA-BB to PC. • Run the TETRA-BB soft front panel (SFP). After selecting the "Exp. 9 - FSK and PSK Modulation & Demodulation" experiment from the menu, TETRA-BB software will provide guidance for the steps of	▪ Ensure that the NI ELVIS III Application Board power button at the top left corner of the unit is OFF (not illuminated). ▪ Carefully plug the Emona Communications board into the NI ELVIS III ensuring that it is fully engaged both front and back. ▪ Ensure that you have connected the NI ELVIS III to the PC using the USB cable and that the PC is turned on.

implementing frequency shift keying modulation, whose block diagram and the required TETRA-BB connections are shown in Figure 8.10.a (steps 0-3 in the TETRA-BB SFP).	▪ Turn on the Application Board *Power* button by pressing it once and confirm that it is illuminated. The LEDs on the board should also be illuminated. If they are not, then switch the unit off immediately and check for connection or insertion errors. ▪ Open the Instrument Launcher software in your browser and select the required instruments.
	▪ Modify the set-up as shown in figure 8.10.b. (But first you have to follow the below 4 steps, then you have to set-up) ▪ Turn the VCO GAIN control knob fully clockwise (maximum). ▪ Connect the Function Generator CH1 output to the VCO module Vin input. Click the Run button on the Function Generator to start generating a signal. ▪ Watching the Scope Channel 1 Frequency measurement, change the Function Generator Gain value until the VCO output frequency is approximately 130kHz. (*) ▪ With the Gain value that you just set, use the Scope to measure the output voltage of Function Generator CH1.
• Adjust following the oscilloscope panel: Channel 0 Voltage Range: 500mV/div Channel 1 Voltage Range: 1V/div Horizontal Time base: 500us/div Trigger: Digital, TRIG Source, Level:0V Horizontal Position (%):50 Probe Attenuation :1x	▪ Adjust following Scope configurations: Channel 1 Voltage Range: 2V/div Channel 2 Voltage Range: 2V/div Horizontal Time base: 100us/div Trigger: Digital Edge, Source: Trig, Rising Probe Attenuation :1x



(a)



(b)

Figure 8.10. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to generate an FSK signal.

<table>
<tr><th>Red Set</th><th>Blue Set</th></tr>
<tr><td>

- Set frequency of the function generator to 10 kHz and amplitude to 5 V. Set modulation type to FM. Adjust the line code to [00] NRZ-L. Set digital signal frequency to 2 kHz.

</td><td>

- Adjust following function generator configurations:
  Channel 1: Custom
  Update Rate: 10 kS/s
  Gain: 1
  Waveform: ECB_positive1V_DC.csv

</td></tr>
<tr><td>6. Observe and draw signals in time domain at FUNC OUT port of the function generator and LINE CODE port of the Sequence Generator 📷,✎.</td><td rowspan="3">

- After set-up, make sure that the micro-switch found under the EX-OR GATE module is set to MUX. Doing so sets the PARALLEL SERIAL/MUX module to MUX mode.
- Connect Scope CH1 to the MUX module output and connect Scope CH2 to the MUX module CLK/DATA input. Confirm that when the CLK/DATA input is at logic-1, the MUX module output is 0V and when the CLK/DATA input is at logic-0, the MUX module output is the DC value that you determined in step (*).

6. Connect Scope CH1 to the VCO SINE output. Confirm that Scope CH1 shows an FSK waveform with 130kHz periods when the MUX CLK/DATA input is at logic-0 and 100kHz periods when the MUX CLK/DATA input is at logic-1 📷,✎.

</td></tr>
<tr><td>

- Apply the settings of DSA given below:
  Input Settings: Channel 0
  Center Frequency: 100000
  Frequency Span: 20000
  Mode: RMS
  Resolution (lines): 3200
  Weighting: Exponential
  Window: 7 Term-B Harris
  # of Averages: 5

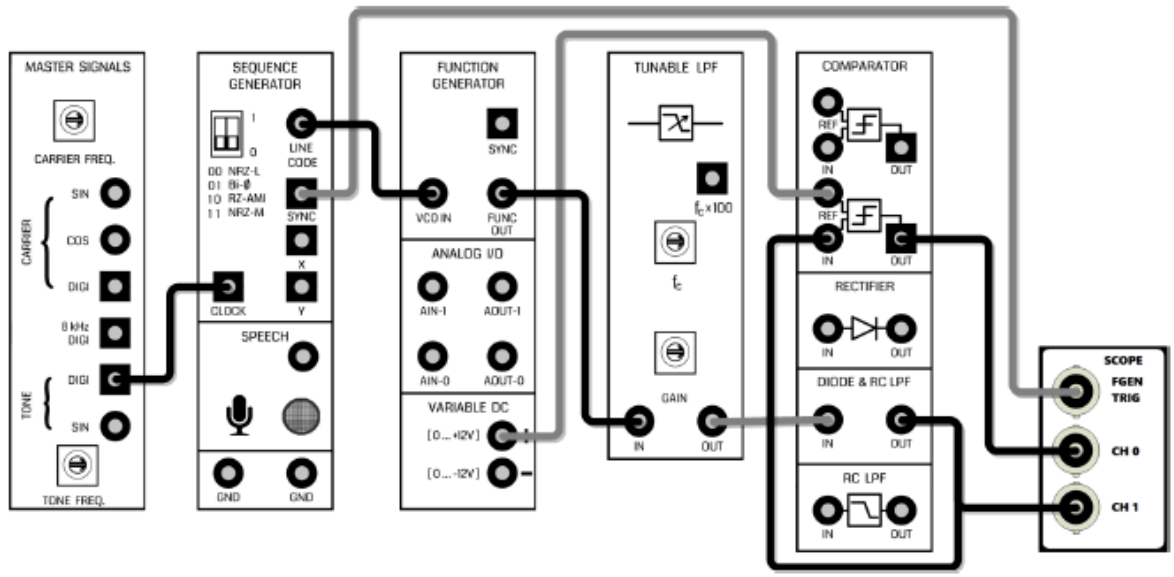
</td></tr>
<tr><td rowspan="2">7. Set the frequency to 100 kHz of the function generator. Stop the oscilloscope. Switch oscilloscope to DSA in TETRA-BB SFP. Observe and draw the frequency spectrum of the FSK modulated signal 📷,✎.</td></tr>
<tr><td>

- Enable the FFT mode of the Oscilloscope instrument. Change the scope's timebase to 2V/div.
- Set the frequency span.
- Select the Hamming window.

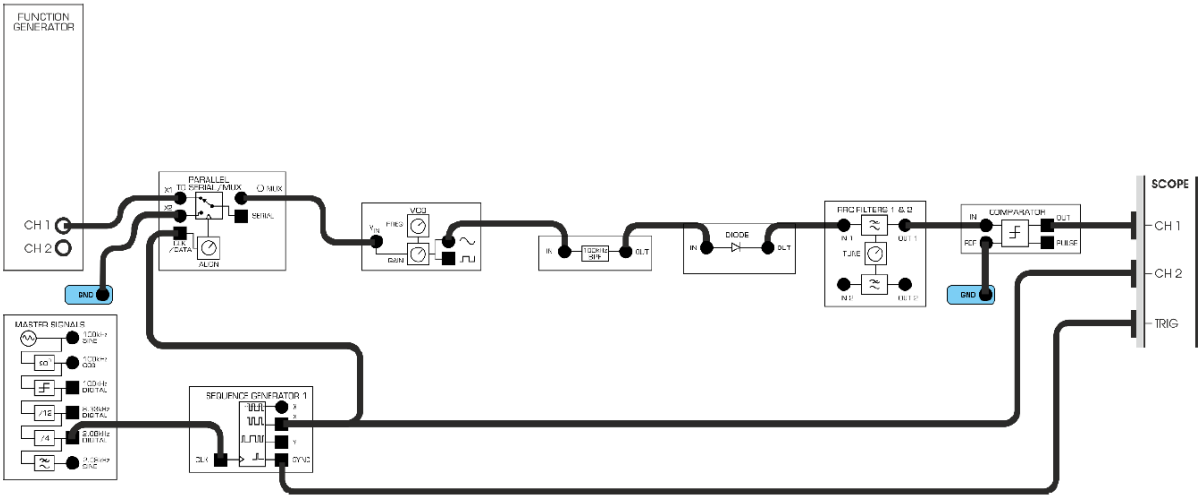
</td></tr>
<tr><td></td><td>7. Stop oscilloscope. Enable the FFT mode of the Oscilloscope instrument. Observe and draw the frequency spectrum of the FSK modulated signal 📷,✎.</td></tr>
</table>

3.4. Demodulating an FSK signal

Red Set	Blue Set
• Please turn on the power of TETRA-BB and ELVIS platform. Then, connect the ELVIS platform to PC (via USB). • Set the control mode of TETRA-BB to PC. • Run the TETRA-BB soft front panel (SFP). After selecting the "Exp. 9 - FSK and PSK Modulation & Demodulation" experiment from the menu, TETRA-BB software will provide guidance for the steps of implementing frequency shift keying modulation, whose block diagram and the required TETRA-BB connections are shown in Figure 8.11.a (steps 4-8 in the TETRA-BB SFP).	▪ Ensure that the NI ELVIS III Application Board power button at the top left corner of the unit is OFF (not illuminated). ▪ Carefully plug the Emona Communications board into the NI ELVIS III ensuring that it is fully engaged both front and back. ▪ Ensure that you have connected the NI ELVIS III to the PC using the USB cable and that the PC is turned on. ▪ Turn on the Application Board *Power* button by pressing it once and confirm that it is illuminated. The LEDs on the board should also be illuminated. If they are not, then switch the unit off immediately and check for connection or insertion errors. ▪ Open the Instrument Launcher software in your browser and select the required instruments. ▪ Modify the set-up as shown in figure 8.11.b.



(a)



(b)

Figure 8.11. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to demodulate an FSK signal.

8. Adjust the cut-off frequency and gain of tunable low pass filter and variable DC voltage to regenerate the input signal. Observe and draw input and demodulated signals in time domain .